MATH1860J Recitation Class 01

Foundations: Ch.01-06

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Outline

1. Introduction
2. Logic
3. Set Theory
4. The Natural Numbers

1. Introduction

Introduction

About me:

- **Shuhan Yu**: Sophomore student majoring in ECE at GC.
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- **Office Hours:**
- **Interest:** Quantitative Trading, Machine Learning, and Geoscience.

About Honors Mathematics Sequence (8 series):

- Emphasize on rigorous mathematical reasoning and proof.
- Write every step of your solution clearly and logically.
- Don't be frightened by "Honor" in the course title. Remember that 5 series is also an honor series.
- Sample exams are of great help. Go through them before exams.

Feel free to interrupt and ask questions if you have any doubt during RCs!

2. Logic

Propositional Logic

Definition

A **statement** (also called a **proposition**) is a declarative sentence that can be unambiguously classified as either *true* or *false*.

- **Logical operations:**
  - Negation (unary operation): $\neg A$
  - Conjunction: $A \wedge B$
  - Disjunction: $A \vee B$
  - Implication: $A \to B$
  - Converse: $B \to A$
  - Equivalence: $A \leftrightarrow B$

  Binary operations return a compound statement.
- **Special propositions:**
  - Tautology: always true
  - Contradiction: always false
  - Vacuous truth.

The definition of logical operations is often given by a **truth table**. Consider the implication $A \to B$. It can be rewritten in several equivalent forms (where $\equiv$ denotes logical equivalence):

$$A \to B \;\equiv\; \neg A \vee B \;\equiv\; \neg(A \wedge \neg B) \;\equiv\; \neg B \to \neg A.$$

A	B	$A \to B$	$\neg A \vee B$	$\neg(A \wedge \neg B)$	$\neg B \to \neg A$
T	T	T	T	T	T
T	F	F	F	F	F
F	T	T	T	T	T
F	F	T	T	T	T

With the basic operation laws established, we can use them to simplify and analyze more complicated propositions.

Remark: (Important for Proof)

- $A \to B \equiv \neg B \to \neg A$: equivalence with the **contrapositive**.
- $A \to B \equiv \neg(A \wedge \neg B)$: useful in **proof by contradiction**.
- $A \to B \equiv \neg A \vee B$: converts the implication into a disjunction, often used in algebraic manipulation.

Important logical equivalences and laws:

- **Commutativity:** $p \vee q \equiv q \vee p$, $p \wedge q \equiv q \wedge p$
- **Associativity:** $(p \vee q) \vee r \equiv p \vee (q \vee r)$
- **Distributivity:**

$$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$
$$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

- **De Morgan's Laws:**

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$
$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Predicate Logic

A **statement** can be expressed as:

$$\textbf{statement} = \textbf{quantifier} + \textbf{predicate (statement frame)}.$$

Definition

A **predicate** is an expression $E(x)$ that becomes a statement when x is replaced by an object from **a specified class** of objects.

- **Universal quantifier** $\forall$: "for all". The expression $\forall x \in X : E(x)$ means "Every object x in X satisfies property E".
- **Existential quantifier** $\exists$: "there exists". The expression $\exists x \in X : E(x)$ means "There exists at least one object x in X with property E".
- **Unique existential** $\exists!$: exactly one such object exists.

In all cases, **the order of the quantifiers is significant.** Thus,

$$\forall x\ \exists y : E(x, y) \quad \not\Leftrightarrow \quad \exists y\ \forall x : E(x, y)$$

- Former, for all x there exists some y such that $E(x, y)$ is true.
- Latter, there exists a fixed y such that $E(x, y)$ is true for all x.

Exercise 01

1. Show that

$$(A \wedge B) \to C \;\equiv\; A \to B \to C \;\equiv\; B \to A \to C.$$

2. Give the negations of the statements:

$$\forall x \in X, \exists y \in Y : E(x, y, z)$$

$$\forall \epsilon > 0, \exists \delta > 0 : |x - a| < \delta \implies |f(x) - f(a)| < \epsilon$$

Remarks:

Using the quantifiers $\exists$ and $\forall$, negation becomes a purely "mechanical" process in which the symbols $\exists$ and $\forall$ (as well as $\wedge$ and $\vee$) are interchanged (without changing the order), and statements are negated.

3. Set Theory

Set Basics and Operations

Sets are collections of objects, called **elements** or **members**.

- Define by Listing: $A = \{1, 2, 3, 4\}$
- Define by Predicate: $A = \{x \in \mathbb{N} \mid 1 \leq x \leq 4\}$
- Set $A = B$ iff $A \subseteq B$ and $B \subseteq A$. (Useful in Proof)

Let $A = \{x \mid P_1(x)\}$ and $B = \{x \mid P_2(x)\}$.

- Union: $A \cup B := \{x \mid P_1(x) \vee P_2(x)\}$
- Intersection: $A \cap B := \{x \mid P_1(x) \wedge P_2(x)\}$
- Difference: $A \setminus B := \{x \mid P_1(x) \wedge \neg P_2(x)\}$.
- Complement: $A^c(\overline{A}) := \mathbb{U} \setminus A := \{x \in \mathbb{U} \mid \neg P_1(x)\}$.

If $A \cap B = \emptyset$, we say that A and B are **disjoint**.

Remark: Distinguish between **Difference**$(A \setminus B)$ and **Quotient**(A/B)

Exercise 02

Verify the following set-theoretic identities:

1. $(A \cup B)^c = A^c \cap B^c$
2. $(A \cap B)^c = A^c \cup B^c$
3. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
5. $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$
6. $(A \cap B) \setminus C = (A \setminus C) \cap (B \setminus C)$
7. $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$
8. $A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

Hint: Use element-wise reasoning and draw a Venn diagram.

Power Set and Cardinality

- **Power Set:** $P(A) = \{X \mid X \subseteq A\}$
- **Cardinality:** $|A|$ is the number of elements in set A
- **Finite Set:** A set with a finite number of elements
- **Infinite Set:** A set with an infinite number of elements
- **Countable Set:** A set that can be put into one-to-one correspondence with the natural numbers
- **Uncountable Set:** A set that cannot be put into one-to-one correspondence with the natural numbers
- **Cardinality of Power Set:** $|P(A)| = 2^{|A|}$ if A is finite

Ordered Pairs and Cartesian Product

- **Ordered Pair:** (a, b)
- **Cartesian Product:** $A \times B = \{(a, b) \mid a \in A, b \in B\}$
- **n-tuple**: generalization of ordered pair, $(a_1, a_2, \ldots, a_n)$
- Construction of ordered pair and natural numbers: $(a, b) = \{\{a\}, \{a, b\}\}; 0 = \{\emptyset\}, 1 = \{\emptyset\}, 2 = \{\emptyset, \{\emptyset\}\}, \ldots$

- Note that $(a, b) = (c, d) \iff a = c \wedge b = d$.
- In general, $A \times B \neq B \times A$.

Definition

A **binary relation** on a set X is a subset $R \subseteq X \times X$. Instead of writing $(x, y) \in R$, we usually write xRy or $x \sim_R y$.

Properties of Relations:

- **Reflexive:** R is reflexive if xRx for all $x \in X$.
- **Transitive:** R is transitive if $(xRy) \wedge (yRz) \implies xRz$.
- **Symmetric:** R is symmetric if $xRy \implies yRx$.

Let $Y \subseteq X$ be nonempty. The **restriction** of R to Y is

$$R_Y := (Y \times Y) \cap R.$$

Clearly, xR_Yy if and only if $x, y \in Y$ and xRy. Usually, we simply write R instead of R_Y when the context is clear.

Equivalence Relation

Definition (Equivalence Relation)

A relation $\sim$ on a set X is called an **equivalence relation** if it is:

- **Reflexive:** $x \sim x$ for all $x \in X$,
- **Symmetric:** $x \sim y \implies y \sim x$,
- **Transitive:** $(x \sim y \wedge y \sim z) \implies x \sim z$.

Definition (Equivalence Class and Quotient Set)

For $x \in X$, the **equivalence class** of x is

$$[x] := \{y \in X \mid y \sim x\}.$$

The set of all equivalence classes of X is

$$X/\sim := \{[x] \mid x \in X\},$$

called X **modulo** $\sim$.

Partition

Definition (Partition)

A **partition** of a set X is a subset $\mathcal{A} \subseteq \mathcal{P}(X) \setminus \{\emptyset\}$ such that:

- For each $x \in X$, there is a unique $A \in \mathcal{A}$ with $x \in A$,
- The elements of $\mathcal{A}$ are pairwise disjoint, and $\bigcup \mathcal{A} = X$.

Proposition

Let $\sim$ be an equivalence relation on X. Then $X/\sim$ is a partition of X.

For all $x \in X$, $x \in [x]$, so $X = \bigcup_{x \in X} [x]$. If $z \in [x] \cap [y]$, then $z \sim x$ and $z \sim y$, hence $x \sim y$, which implies $[x] = [y]$. Thus, equivalence classes are either identical or disjoint.

Order Relations

Definition (Partial Order)

A relation $\leq$ on a set X is a **partial order** if it is:

- **Reflexive:** $x \leq x$ for all $x \in X$,
- **Transitive:** $x \leq y \wedge y \leq z \implies x \leq z$,
- **Antisymmetric:** $x \leq y \wedge y \leq x \implies x = y$.

The pair $(X, \leq)$ is called a **partially ordered set**. If the order is clear from context, we simply write X.

Definition (Total Order)

If, in addition,

$$\forall x, y \in X : (x \leq y) \vee (y \leq x),$$

then $\leq$ is a **total order**, and $(X, \leq)$ is called a **totally ordered set**.

Order Relations

Useful notations: $x \geq y \;:\Longleftrightarrow\; y \leq x$, $x < y \;:\Longleftrightarrow\; x \leq y \wedge x \neq y$, $x > y \;:\Longleftrightarrow\; y < x$.

Remark

If X is **totally ordered**, then for each $x, y \in X$, exactly one of $x < y$, $x = y$, $x > y$ holds.

Remark

If X is **partially ordered** but not totally ordered, there exist elements $x, y \in X$ which are **incomparable**, i.e., neither $x \leq y$ nor $y \leq x$ holds.

Definition (Upper and Lower Bounds)

Let $(X, \leq)$ be a partially ordered set and $A \subseteq X$ nonempty.

- An element $s \in X$ is an **upper bound** of A if $a \leq s$ for all $a \in A$.
- An element $s \in X$ is a **lower bound** of A if $a \geq s$ for all $a \in A$.
- A is **bounded above** if it has an upper bound, **bounded below** if it has a lower bound, and **bounded** if both.

Definition (Bounded and Unbounded Sets)

A set $U \subset \mathbb{Q}$ is **bounded** if there exists a constant $c \in \mathbb{Q}$ such that

$$|x| \leq c \quad \text{for all } x \in U.$$

If no such c exists, we say that U is **unbounded**.
Numbers c_1 and c_2 such that

$$c_1 \leq x \leq c_2 \quad \text{for all } x \in U$$

are called **lower and upper bounds** of U, respectively.
A set $U \subset \mathbb{Q}$ is **bounded above** if it has an upper bound, and **bounded below** if it has a lower bound.

Maximum and Minimum

Definition (Maximum and Minimum)

Let $U \subset \mathbb{Q}$.

- $x_1 \in U$ is the **minimum** of U, denoted $x_1 = \min U$, if

$$x_1 \leq x \quad \text{for all } x \in U.$$

- $x_2 \in U$ is the **maximum** of U, denoted $x_2 = \max U$, if

$$x_2 \geq x \quad \text{for all } x \in U.$$

A set not bounded below has no minimum, and a set not bounded above has no maximum. Even a bounded set may fail to have a maximum or minimum.

Definition (Supremum and Infimum)

Let $U \subset \mathbb{Q}$.

- An upper bound $c_2 \in \mathbb{Q}$ of U is the **least upper bound** or **supremum**, denoted $c_2 = \sup U$, if no $c \in \mathbb{Q}$ with $c < c_2$ is an upper bound.
- A lower bound $c_1 \in \mathbb{Q}$ of U is the **greatest lower bound** or **infimum**, denoted $c_1 = \inf U$, if no $c \in \mathbb{Q}$ with $c > c_1$ is a lower bound.

Naive Set Theory: You can define a set by any property. A set is any collection of distinct objects. **Problem: Russell's Paradox** arises when considering the set of all sets that do not contain themselves:

$$S = \{x \mid x \text{ is a set and } x \notin x\}$$

Question: Is $S \in S$?

- If $S \in S$, then by definition $S \notin S$.
- If $S \notin S$, then by definition $S \in S$.

Conclusion: Contradiction arises. Not every property can define a set freely. $\Rightarrow$ **Implication:** Motivates axiomatic set theory (e.g., Zermelo-Fraenkel Set Theory, ZFC) to avoid such paradoxes.

Cantor's Paradox ()

Cantor's Theorem: For any set A,

$$|A| < |P(A)|,$$

where $P(A)$ is the power set of A, i.e., the set of all subsets of A.
Consider: The "set of all sets" U.

- By definition, U contains every set, so $P(U) \subseteq U$.
- By Cantor's Theorem, $|U| < |P(U)|$.

Contradiction: $|U| \geq |P(U)|$ from definition, but $|U| < |P(U)|$ by Cantor.
Conclusion: There is no "set of all sets". This prevents the paradox.

Zermelo-Fraenkel Set Theory (ZF)

1. **Axiom of Extensionality:** Two sets are equal if they have the same elements.
2. **Axiom of Empty Set:** There exists a set with no elements, denoted $\emptyset$.
3. **Axiom of Pairing:** For any sets a and b, there exists a set $\{a, b\}$.
4. **Axiom of Union:** For any set of sets A, there exists a set $\bigcup A$ containing all elements of elements of A.
5. **Axiom of Power Set:** For any set A, there exists a set $P(A)$ of all subsets of A.
6. **Axiom of Infinity:** There exists a set containing $\emptyset$ and closed under successor.
7. **Axiom of Replacement:** The image of a set under any definable function is also a set.
8. **Axiom of Separation (Specification):** For any set A and property P, $\{x \in A \mid P(x)\}$ is a set.
9. **Axiom of Regularity (Foundation):** Every non-empty set A contains an element disjoint from A.

Exercise 03

Let A, B, C be three statements. Use truth table to prove that

$$(A \vee B) \wedge C \equiv (A \wedge C) \vee (B \wedge C)$$

Let A, B, C be three sets. Prove that (may use Venn Diagram)

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$$

4. The Natural Numbers

Definition (Internal and External Composition Laws)

Let S and S' be two sets.

1. An **internal composition law** $\circ$ is a map

$$S \times S \to S, \quad (x, y) \mapsto x \circ y.$$

2. An **external composition law** $*$ is a map

$$S' \times S \to S, \quad (\alpha, x) \mapsto \alpha * x.$$

Natural Numbers from ZF Set Theory

- By the **Axiom of Infinity**, there exists a set I such that:
  - $\emptyset \in I$
  - For each $x \in I$, $S(x) := x \cup \{x\} \in I$
- Define natural numbers recursively:

$$0 := \emptyset, \quad 1 := S(0) = \{\emptyset\}, \quad 2 := S(1) = \{\emptyset, \{\emptyset\}\}, \ldots$$

- The set of natural numbers $\mathbb{N}$ is the smallest set containing 0 and closed under S.

Recursive Definition of Addition on $\mathbb{N}$

- **Base case:**

$$\forall n \in \mathbb{N}, \quad n + 0 := n$$

- **Successor step:**

$$\forall n, m \in \mathbb{N}, \quad n + S(m) := S(n + m)$$

- *Explanation:* $S(m) = m \cup \{m\}$ is the successor of m, so addition recursively "adds 1" using the successor operation.

Recursive Definition of Multiplication on $\mathbb{N}$

- **Base case:**
$$\forall n \in \mathbb{N}, \quad n \cdot 0 := 0$$
- **Successor step:**
$$\forall n, m \in \mathbb{N}, \quad n \cdot S(m) := (n \cdot m) + n$$
- *Explanation:* Multiplication is defined as repeated addition, consistent with the intuitive meaning of natural numbers.

Definition (Principle of Mathematical Induction)

Let $A(n)$ be a statement depending on $n \in \mathbb{N}$ and let $n_0 \in \mathbb{N}$. To prove $A(n)$ for all $n \geq n_0$, it suffices to verify:

1. **Base Case:** $A(n_0)$ is true.
2. **Inductive Step:** For all $n \geq n_0$, if $A(n)$ is true, then $A(n+1)$ is true:

$$\forall n \in \mathbb{N},\ n \geq n_0 : \quad A(n) \implies A(n+1).$$

Conclusion: If both conditions hold, $A(n)$ is true for all $n \geq n_0$.
Remark: The principle of induction is essentially part of the definition of the natural numbers $\mathbb{N}$.

Exercise

Suppose M is a finite set of men.
Proposition: For any $a, b \in M$, $a = b$.
Proof (by induction on $|M|$):

1. **Base Case:** If $|M| = 1$, the claim is trivially true.
2. **Inductive Step:** Assume true for n men. Let $|M| = n + 1$ and $a, b \in M$. Define $M_a = M \setminus \{a\}$, $M_b = M \setminus \{b\}$. Take $c \in M_a \cap M_b$. Then $a = c,\ b = c \implies a = b$.

Question: Where is the flaw in this proof?

References

- Horst Hohberger, *math186_all_lecture_slides (2026)*
- Wenxin Chen, *MATH1860J RC*, 25 Fall
- Leqi Tang, *MATH1860J RC*, 25 Fall

Thanks for Listening!

Any questions? Please feel free to ask!